Waleed Abdulqawi Al-Asad

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Summary

Data scientist with three years of client-facing work in banking, public finance, healthcare, education and Hajj operations, building models that run inside the client's environment. Author of autoLRP, an open-source library for explaining PyTorch models.

Experience

Mozn AI Sep 2023 – Present

Data Scientist III, Forward Deployed *Riyadh, SA*

Production deliveries

Client: **Alrajhi Bank: anti-fraud**

- Turned card-fraud rule discovery into a local-search optimization over a multi-objective score, replacing a manual process that took weeks with one that takes minutes. The deployed rules cut card fraud by about 25%.
- Deployed real-time mule-cashout controls: about 3M SAR/month prevented at 50% precision.
- Built the scheduled pipelines that pull billions of transactions, compute features and store them for fraud analysis.

Client: **NCGR (Ministry of Finance): budget validation**

- Fine-tuned embedding models on budget-allocation descriptions so comptrollers can cross-check allocations against each other: 70% Hits@5, against 28% for the off-the-shelf embeddings.
- Built a two-stage OCR pipeline that reads handwritten Arabic digits on quantity bills into structured data.

Client: **SCFHS (Health Specialties Commission): Digital Mentor**

- Modeled the rank-based nomination process as a Bayesian model in PyMC, fitted by MCMC: it infers per-program parameters and simulates each candidate's chance of nomination. F1 about 0.84, against about 0.39 for the baseline.
- Tested for gender and nationality bias (gender-flip tests, sparse-nationality checks) and designed outcome-based KPIs for universities and training centers; presented to steering committees and endorsed by the Deputy Secretary.

Client: **NeLC (National eLearning Center): learner analytics**

- Led the development of 30+ analytical and ML models for a platform with 200,000+ learners: skill tracking across courses, course recommendation, and a learning-pathway optimizer (shortest path over courses, with difficulty, rating and skill gain as edge weights).
- Moved inference into Airflow jobs and served cached outputs: API response time fell from about 5 s to 150 ms.

Client: **MOHU (Ministry of Hajj and Umrah): season analytics**

- Analyzed 100M+ records from past seasons in Polars and found fraudulent campaigns, price manipulation and country-level adherence gaps; the reports informed the Deputy Minister's enforcement and operations decisions.

Proofs of concept

Client: **MEWA (Ministry of Environment, Water and Agriculture): public-services assistant**

- Built a RAG conversational assistant over 300+ public services, served with vLLM on GCP; a custom tree-structured retriever scored 96% on the evaluation set against 82% for an OpenAI-embedding baseline.

Client: **SCFHS: SMLE question generator**

- Benchmarked several LLMs against medical practitioners on exam-question quality, then built a generator for SMLE-standard questions grounded in medical textbooks (RAG), served with vLLM on GCP at 2.5× the throughput of the baseline. The POC won a second-phase engagement.

Client: **SAMA (Central Bank): regulatory compliance auditing**

- Designed an agentic LLM pipeline that reads SAMA's policy corpus into enforcement criteria, writes validation code against an audited entity's data, and produces a compliance report listing the gaps.

Intelmatix Jun 2022 – Aug 2022

Machine Learning Intern *Riyadh, SA*

- Built UMAP-based clustering pipelines that segment geographical data for downstream models.
- Ran market and competitor analysis to shortlist and rank ML use cases against business goals.

Education

King Fahd University of Petroleum and Minerals (KFUPM) Sep 2018 – Jun 2023

B.Sc. Software Engineering, First Honor; AI & Machine Learning concentration *GPA: 3.96/4.00*

Open Source & Courses

autoLRP: model-agnostic explainability library for PyTorch (author), github.com/Wa-lead/autoLRP 2025 – 2026

- Layer-wise Relevance Propagation with the propagation rules written at the operation level rather than per layer or architecture, which gives full coverage of any PyTorch model without a rewrite; tested across model families (CNNs, ViT, LLMs, Mamba, world models) with full coverage of each. MIT-licensed, on PyPI as `autolrp`.

SimEngine: text-similarity toolkit, github.com/Wa-lead/SimEngine

- Toolkit for text-similarity tasks with BERT, FastText and NER-based methods built in and an interface for adding custom ones.

Machine Learning Compilation (course) 2025

- Tensor program abstraction, automatic program optimization, GPU acceleration, computational-graph optimization.

PyTorch from scratch, github.com/Wa-lead/needle-pytorch-from-scratch 2024

- Built a PyTorch-like framework following Carnegie Mellon's Deep Learning Systems course: autodiff over a computation graph, the core operations, a custom C++ backend, a layers API, and a Transformer language model on top of it.

Nand2Tetris (course) 2025

- Built a computer from logic gates up: CPU and memory, assembler, virtual machine, compiler for an object-oriented language, and basic OS components.

Research

Research Assistant, Dr. Moayad Alnammi Sep 2022 – Mar 2023

SDAIA-KFUPM Joint Research Center for AI

- Compared machine-learning and deep-learning methods for ligand-based virtual screening, with a focus on molecule featurization. Unpublished.

Research Assistant, Dr. Wessam Mesbah Jan 2022 – Oct 2022

KFUPM

- Developed a greedy partitioning algorithm for placing firewalls in large IoT networks to block botnet formation; it needs 30% fewer firewalls than clustering approaches, consistently across network sizes. Unpublished.

Skills

LLM systems: RAG, agentic pipelines, embedding fine-tuning, LLM evaluation and benchmarking, vLLM serving
ML: PyTorch, deep learning, NLP, computer vision, Bayesian modeling (PyMC), optimization, explainability
Engineering: Python, SQL, SAS, Polars, Airflow, Docker, GCP
Languages: Arabic (native), English (TOEFL iBT 104)

Communities

BCG Jeel Tamooh (2022): top 100 students nationally; trained by BCG leadership.

Project1932 LittleSib (2021–22): one-to-one mentorship by a leading AI figure in the region.